Biology

Class 9 (English Medium)

Solved Exercises — Sindh Textbook Board

9 lessons — 352 questions

حیاتیات جو تعارف

Question: 1. (i) What is a group of organisms of the same species living together in one place called?

Answer: (d) Population. Organisms of the same species living together in one place form a population.

Question: 1. (ii) What is the breeding of good fish and fish production called?

Answer: (a) Fisheries. The work of breeding good fish and increasing fish production is called fisheries.

Question: 1. (iii) With the help of which science can the age of all organic life be known?

Answer: (b) Palaeontology (the science of fossils). The age of past life is known through the study of fossils.

Question: 1. (v) Which of the following is the wrong statement?

Answer: (a) is the wrong statement. The six basic elements are carbon, hydrogen, oxygen, nitrogen, phosphorus and sulphur; this statement leaves out sulphur, so it is incomplete.

Question: 1. (vi) What is the science of the treatment of diseases called?

Answer: (d) Both (b) and (c). Diseases are treated through both the science of medicines (pharmacology) and surgery.

Question: 1. (vii) What do similar cells together make?

Answer: (c) Tissue. Cells of the same kind together make a tissue.

Question: 1. (viii) What is the scientific name of the frog?

Answer: (b) Rana tigrina. The scientific name of the frog is Rana tigrina.

Question: 1. (ix) What is the correct order of biological organisation?

Answer: (c) Atom → molecule → cell → tissue → organ. The order of

organisation runs from small to large: atom, molecule, cell, tissue, organ.

Question: 1. (x) Volvox is a multicellular body of which kind of algae?

Answer: (a) Green algae. Volvox is a colony of green algae.

Question: 2. (ii) Dividing different organisms into different groups or kingdoms is called .....

Answer: Classification. Dividing organisms into groups on the basis of their characteristics is called classification.

Question: 2. (iii) The part of agriculture meant for new plant varieties and the growth of fruits is called .....

Answer: Horticulture. In horticulture, new plant varieties are produced and the growth of fruits is cared for.

Question: 2. (iv) The elements necessary for life are found in .....

Answer: All living things. The elements necessary for life are found in all living things.

Question: 2. (v) Members of different species living together at a particular place are called .....

Answer: Community. Members of different species living at one place together form a community.

Question: 2. (vii) The part of the Earth where life is found is called .....

Answer: Biosphere. The part of the Earth where life is found is the biosphere.

Question: 2. (viii) The basis of life rests on .....

Answer: Protoplasm. Protoplasm is the chemical basis of life.

Question: 2. (ix) Fish is a good source of .....

Answer: Protein. Fish contains a lot of protein.

Question: 2. (x) Radio-dating and carbon dating are used to find out the ..... of fossils.

Answer: Age. Radio-dating and carbon dating are used to find out the age

of fossils.

Question: 3. (i) What is anatomy?

Answer: Anatomy is the science in which the internal structure of the bodies of organisms is studied.

Question: 3. (ii) What is histology?

Answer: Histology is the science in which the structure of tissues is studied.

Question: 3. (iii) What is immunology?

Answer: Immunology is the science in which the body's immune system and protection against diseases is studied.

Question: 3. (iv) What is pharmacology?

Answer: Pharmacology is the science of medicines; in it the effect and use of medicines are studied.

Question: 3. (v) What is entomology?

Answer: Entomology is the science of insects.

Question: 3. (vi) What is biometry?

Answer: Biometry is the science in which biological measurement and statistics are used.

Question: 3. (vii) What is biogeography?

Answer: Biogeography is the science in which the geographical distribution of organisms on the Earth is studied.

Question: 3. (viii) What is surgery?

Answer: Surgery is the science in which diseases are treated by means of operations.

Question: 3. (ix) What is the science of animal husbandry?

Answer: The science of animal husbandry is the science in which rearing animals, caring for them and gaining benefit from them are studied.

Question: 3. (x) What are bio-elements?

Answer: Bio-elements are the elements necessary for life, such as carbon, hydrogen, oxygen, nitrogen, phosphorus and sulphur.

Question: 4. (i) State the difference between colonial organisation and multicellular organisation.

Answer: In colonial organisation many cells live together but each cell does its own work, as in Volvox. In multicellular organisation cells together form tissues, organs and systems, and the work is divided.

Question: 4. (ii) State the difference between agriculture and horticulture.

Answer: In agriculture, crops are grown. Horticulture is the part of agriculture in which new plant varieties, fruits and flowers are grown.

Question: 5. (i) Why is biology called a multidisciplinary subject?

Answer: Biology is called a multidisciplinary subject because it is related to many other sciences, such as physics, chemistry, mathematics and geography, and it has many branches.

Question: 5. (ii) How does biology serve farmers?

Answer: Biology serves farmers by increasing crop production, producing new plant varieties, controlling pests and improving animal rearing.

Question: 5. (iii) How is a species recognised?

Answer: A species is recognised when its members can interbreed and produce fertile offspring.

Question: 5. (iv) How is a population different from a community?

Answer: A population is a group of members of one species, whereas in a community members of different species live together.

Question: 5. (v) How are new plant varieties produced?

Answer: New plant varieties are produced through selection, grafting and genetic methods.

Question: 6. (i) Mention the contribution of Muslim scientists in biology.

Answer: Muslim scientists made a great contribution to biology. Jabir bin

Hayyan worked in chemistry, Ibn Sina in medicine, Al-Razi in skin diseases and smallpox-measles, and Al-Biruni and others in botany and zoology. Scientists of Europe also benefited from their books.

Question: 6. (ii) Describe the relationship of biology with other branches.

Answer: Biology is related to physics (biophysics), chemistry (biochemistry), mathematics (biometry), geography (biogeography), and medicine, agriculture and other fields. That is why biology is called a multidisciplinary subject.

Question: 6. (iii) Describe the levels of organisation.

Answer: The levels of organisation run from small to large: atom → molecule → cell → tissue → organ → system → organism → population → community → ecosystem → biosphere. Each larger level is made of the smaller levels.

حياتياتي مسئلو حل ڪرڻ

Question: 1. (i) What is the correct sequence of the biological method?

Answer: (c) Hypothesis → reasoning → theory → law. The sequence of the biological method is: problem, observation, hypothesis, reasoning, experiment, data collection, conclusion, theory and law.

Question: 1. (ii) Which of the following is the odd one out?

Answer: (d) Ratio. Theory, law and hypothesis are parts of the biological method, whereas ratio is a concept of mathematics.

Question: 1. (iii) What is the science that studies biological systems by mathematical methods called?

Answer: (b) Biomathematics. In biomathematics, biological systems are studied by mathematical methods.

Question: 1. (iv) The biological method includes all of the following except one. Which one?

Answer: (d) Proportion. The biological method includes problem, observation, hypothesis, experiment and data collection; proportion is not a part of it.

Question: 1. (v) What is scientific reasoning from the specific to the general called?

Answer: (a) Inductive reasoning. In inductive reasoning a general conclusion is drawn from specific observations.

Question: 1. (vi) What is used in quantitative observation?

Answer: (b) Instruments. In quantitative observation, instruments are used for numbers and measurement.

Question: 1. (vii) What is an equation in which two ratios are equal called?

Answer: (b) Proportion. When two ratios are equal, that equation is a

proportion.

Question: 1. (viii) How is the relationship between two variables shown?

Answer: (c) Graph. The relationship between two variables is shown by means of a graph.

Question: 1. (ix) What is a hypothesis?

Answer: (b) A testable statement which is checked and confirmed. A hypothesis is a statement that can be tested through experiment.

Question: 1. (x) Which method is most useful for arranging data?

Answer: (d) Both (a) and (b). Both tables and graphs are useful for arranging data.

Question: 2. (i) Problems on which biologists do not agree are called .....

Answer: Biological problems. Problems on which scientists debate are biological problems.

Question: 2. (ii) The solution of a biological problem begins with .....

Answer: Observation. The solution of a biological problem begins with observation.

Question: 2. (iii) An important part of the scientific method is .....

Answer: Hypothesis. Hypothesis is an important part of the scientific method.

Question: 2. (iv) Scientific reasoning based on "if-then" statements is called .....

Answer: Deductive reasoning. Deductive reasoning is based on "if-then" statements.

Question: 2. (v) The last step of the scientific method shows .....

Answer: Theory. The last step of the scientific method is theory, which is formed from tested conclusions.

Question: 2. (vi) The unchanging answer to a uniform or constant natural fact is .....

Answer: Law. A scientific law is a statement of a uniform and constant natural fact.

Question: 2. (vii) Once data is in hand, you will need ..... analysis.

Answer: Mathematical. Once data is in hand, its mathematical analysis is needed.

Question: 2. (viii) An equation in which two ratios are equal is called .....

Answer: Proportion. An equation of two equal ratios is called a proportion.

Question: 2. (ix) Ratio is the comparison of ..... quantities.

Answer: Two. Ratio is the comparison of two quantities.

Question: 2. (x) The cause of malaria is .....

Answer: Plasmodium. Malaria is caused by the organism called Plasmodium.

Question: 3. (i) What is ratio?

Answer: Ratio is the comparison of two quantities, such as two divided by four.

Question: 3. (ii) What is the biological method?

Answer: The biological method is the way of solving biological problems by the scientific method: problem, observation, hypothesis, reasoning, experiment, data collection, conclusion, theory and law.

Question: 3. (iii) What is a graph?

Answer: A graph is a way of showing data in picture form, in which the relationship between two variables is shown.

Question: 3. (iv) What is a hypothesis?

Answer: A hypothesis is a testable statement that can be checked through experiment.

Question: 3. (v) What is a law?

Answer: A law is a statement of a uniform and constant natural fact, which has been tested again and again.

Question: 3. (vi) What is inductive reasoning?

Answer: Inductive reasoning is reasoning in which a general conclusion is drawn from specific observations.

Question: 3. (vii) What is a derived conclusion?

Answer: A derived conclusion is the conclusion drawn from experiment and analysis of data.

Question: 3. (viii) What is proportion?

Answer: Proportion is the equation in which two ratios are equal.

Question: 3. (ix) What is observation?

Answer: Observation is looking at something carefully and collecting information about it.

Question: 3. (x) What is a mathematical model?

Answer: A mathematical model is representing a biological system by mathematical equations and numbers.

Question: 4. (i) State the difference between theory and law.

Answer: A theory is a tested explanation of a natural process, whereas a law is a statement of a uniform and constant natural fact. A theory tells "why", a law tells "what happens".

Question: 4. (ii) State the difference between inductive and deductive reasoning.

Answer: In inductive reasoning a general conclusion is drawn from specific observations. In deductive reasoning a specific conclusion is drawn from a general principle, usually with "if-then" statements.

Question: 5. (i) Why is a theory a most reliable scientific explanation?

Answer: A theory is a most reliable scientific explanation because it has been tested again and again by experiments and it explains many observations.

Question: 5. (ii) Why does biological science need mathematical models?

Answer: Biological science needs mathematical models so that data can be analysed, patterns can be understood and predictions can be made.

Question: 5. (iii) Make a chart or diagram showing the stages of the biological method.

Answer: Stages of the biological method: problem → observation → hypothesis → reasoning (inductive and deductive) → experiment → data collection → conclusion → theory → law.

Question: 5. (iv) Why are tables and graphs needed for organising data?

Answer: Tables and graphs are needed because with them data can be seen in order, compared, and patterns can be easily understood.

Question: 5. (v) Why is an experiment needed for a theory?

Answer: An experiment is needed for a theory so that the hypothesis can be tested; without experiment a hypothesis cannot become a theory.

حياتياتي فرق

Question: 1. (i) Which of the following is the correct way for the scientific name of organisms?

Answer: (d) Canis lupus. A scientific name is written by the binomial method: the first name is of the genus and the second is of the species. The scientific name of the dog is Canis lupus.

Question: 1. (ii) Choose the wrongly matched pair.

Answer: (d) Animalia → Amoeba. Amoeba is not a member of the animal kingdom but of Protista, so this pair is wrong.

Question: 1. (iii) Identify the correct order of classification of organisms.

Answer: (b) Kingdom → phylum → class → order → family → genus → species. The order of classification runs from large to small: kingdom, phylum, class, order, family, genus, species.

Question: 1. (iv) In classification, all except one are involved. Which is the one?

Answer: (a) Analogous structures. Classification is based on homologous structures; analogous structures are not a basis for classification.

Question: 1. (v) Choose the rank in the classification order that brings all the others in the list into one circle.

Answer: (d) IV (Class). Class is the largest rank, so it contains order, genus and species all within it.

Question: 1. (vi) In the four-kingdom classification, Kingdom Metaphyta includes all of the following except one. Which is the one?

Answer: (a) Algae. Kingdom Metaphyta (plants) includes bryophyta, gymnosperms and angiosperms; algae are not included in it.

Question: 1. (vii) In the five-kingdom system, where are viruses placed?

Answer: (d) None of these. Viruses are not placed in any of the five kingdoms, because they are not cellular.

Question: 1. (viii) What is the biological name of the cat?

Answer: (a) Felis catus. The scientific name of the cat is Felis catus.

Question: 1. (ix) Members of which kingdom are all heterotrophic and have a chitin cell wall?

Answer: (d) Fungi. Members of fungi are all heterotrophic and their cell wall is made of chitin.

Question: 1. (x) What affects biodiversity?

Answer: (d) I, II and III. Biodiversity is affected by deforestation, environmental damage and illegal hunting, all three.

Question: 2. (i) The differences present among species are called .....

Answer: Biodiversity. The differences present among species are called biodiversity.

Question: 2. (ii) A species is given a ..... scientific name.

Answer: Double (binomial). Every species is given a double scientific name: the name of the genus and the name of the species.

Question: 2. (iii) Structures with different functions (physiology) have the same internal ..... structure.

Answer: Homologous. Homologous structures have the same internal structure, even though their functions are different.

Question: 2. (iv) The science of classification is called .....

Answer: Taxonomy. The science of classification is called taxonomy.

Question: 2. (v) Many protists are .....

Answer: Unicellular. Many protists are unicellular organisms.

Question: 2. (vi) Organisms without chlorophyll that take food by absorption are .....

Answer: Fungi. Fungi have no chlorophyll and they take food by

absorption.

Question: 2. (vii) A non-cellular parasite is .....

Answer: Virus. A virus is a non-cellular parasite.

Question: 2. (viii) The scientific name of any organism is written in ..... letters.

Answer: Slanting (italic). A scientific name is written in slanting (italic) letters.

Question: 2. (x) Animals which face the danger of permanent extinction are called .....

Answer: Endangered species. Such animals are called endangered species.

Question: 3. (i) What is analogous?

Answer: Analogous structures are structures that do the same work but have different internal structure.

Question: 3. (ii) What is classification?

Answer: Classification is dividing organisms into groups on the basis of their characteristics.

Question: 3. (iii) What is a species?

Answer: A species is the group of organisms whose members can interbreed and produce fertile offspring.

Question: 3. (iv) What is a family?

Answer: Family is a rank of classification which is smaller than order and larger than genus.

Question: 3. (v) What are Metazoa?

Answer: Metazoa are multicellular animals.

Question: 3. (vi) What is mycelium?

Answer: Mycelium is the network of hyphae (threads) of fungi.

Question: 3. (vii) What are hyphae?

Answer: Hyphae are the thin threads of fungi which together make

mycelium.

Question: 3. (viii) What is a genus?

Answer: Genus is a rank of classification which is smaller than family and larger than species; it includes closely related species.

Question: 3. (ix) What is an endangered species?

Answer: An endangered species is a species which faces the danger of permanent extinction.

Question: 3. (x) What is a kingdom?

Answer: Kingdom is the largest rank of classification, such as Monera, Protista, Fungi, Plantae and Animalia.

Question: 4. (i) State the difference between the plant kingdom and the animal kingdom.

Answer: Members of the plant kingdom make their own food (autotrophic) and their cells contain chlorophyll. Members of the animal kingdom depend on others for food (heterotrophic) and they have no chlorophyll.

Question: 4. (ii) State the difference between Monera and Protista.

Answer: Members of Monera are prokaryotic, meaning they have no true nucleus. Members of Protista are eukaryotic, meaning they have a true nucleus.

Question: 4. (iii) State the difference between Fungi and Plantae.

Answer: Fungi have no chlorophyll and are heterotrophic; Plantae have chlorophyll and are autotrophic. The cell wall of fungi is made of chitin, the wall of Plantae is made of cellulose.

Question: 5. (i) Why is a scientific name necessary?

Answer: A scientific name is necessary so that the same organism is known by the same name everywhere in the world and there is no confusion.

Question: 5. (ii) How were organisms classified into five kingdoms?

Answer: Organisms were classified into five kingdoms on the basis of their cell structure, method of obtaining food and body structure: Monera, Protista, Fungi, Plantae and Animalia.

Question: 5. (iii) Why has the virus not been placed in any kingdom?

Answer: The virus has not been placed in any kingdom because it is not cellular; it is neither living nor dead, but in between the two.

Question: 5. (iv) Make a chart showing the classification of three kingdoms.

Answer: Classification of three kingdoms: (1) Protista — unicellular eukaryotic organisms, (2) Plantae — plants, (3) Animalia — animals.

Question: 5. (v) Why has Amoeba not been placed in the animal kingdom?

Answer: Amoeba has not been placed in the animal kingdom because it is unicellular and has no tissues and organs; it is a member of Protista.

Question: 5. (vi) Why are cyanobacteria placed in Monera?

Answer: Cyanobacteria are placed in Monera because they are prokaryotic, meaning they have no true nucleus.

Question: 6. (i) Explain the classification of five kingdoms.

Answer: There are five kingdoms: (1) Monera — prokaryotic, unicellular, such as bacteria. (2) Protista — eukaryotic, mostly unicellular, such as Amoeba. (3) Fungi — eukaryotic, heterotrophic, chitin wall, such as mushroom. (4) Plantae — eukaryotic, autotrophic, cellulose wall, such as plants. (5) Animalia — eukaryotic, heterotrophic, multicellular, such as animals.

Question: 6. (ii) What is taxonomic hierarchy? State the purpose of classification.

Answer: Taxonomic hierarchy is the order of the ranks of classification: kingdom → phylum → class → order → family → genus → species. The purpose of classification is to identify organisms, bring uniformity in their

names, and understand the relationships among them.

Question: 6. (iii) Describe the effect of deforestation on biodiversity.

Answer: Deforestation destroys the homes of organisms, reduces the number of species, and many species fall into danger. That is why biodiversity decreases.

جیوگھرڑا اوچا

Question: 1. (i) What is responsible for the greater resolution of the electron microscope?

Answer: (b) The short wavelength of the electron beam. In the electron microscope the wavelength of the electron beam is short, so its resolution is greater.

Question: 1. (ii) What are the functions of the endoplasmic reticulum?

Answer: (d) Synthesis of proteins. The rough endoplasmic reticulum makes proteins, and the smooth endoplasmic reticulum makes lipids and steroids.

Question: 1. (iv) Which process drives the movement of things into and out of the cell?

Answer: (d) I, II and III. Osmosis, diffusion and active transport all three cause the movement of substances into and out of the cell.

Question: 1. (v) In cell theory, all except one are postulates. Which is the one?

Answer: (b) The material of the cell membrane has no hereditary material. The postulates of cell theory are: all living things are made of cells, the cell is the basic unit of life, and new cells arise from pre-existing cells.

Question: 1. (vi) What is the secondary cell wall made of?

Answer: (c) Cellulose and lignin. The secondary cell wall is made of cellulose and lignin.

Question: 1. (vii) Choose the odd one out.

Answer: (a) Active transport. Diffusion, facilitated diffusion and osmosis are passive processes in which no energy is spent; active transport uses energy.

Question: 1. (viii) What is the correct sequence for a protein to reach the

Golgi body?

Answer: (b) Ribosome → ER → Golgi body → lysosome. A protein is made in the ribosome, then goes from the endoplasmic reticulum to the Golgi body and then to the lysosome.

Question: 1. (ix) Which small organelles from animal cells help in intracellular digestion?

Answer: (a) Lysosomes. Lysosomes contain digestive enzymes which help in digestion inside the cell.

Question: 1. (x) Choose the mismatched pair.

Answer: (c) Ribosome → synthesis of steroids. Ribosomes make proteins, not steroids; steroids are made in the smooth endoplasmic reticulum.

Question: 2. (i) Microscopes are instruments in which the visible image is .....

Answer: Magnified. In a microscope the visible image is magnified.

Question: 2. (ii) In microscopy the smallest distance between two points is called .....

Answer: Resolution. The smallest distance between two points is called resolution.

Question: 2. (iii) The magnification of a light microscope is about ....., with the combination of lenses.

Answer: 1000 times. In a light microscope the combination of the eyepiece lens and the objective lens gives a magnification of about 1000 times.

Question: 2. (iv) Ordinary light has a longer wavelength than electrons, so the electron microscope gives ..... images.

Answer: Clearer. The electron microscope gives clearer images than an ordinary microscope.

Question: 2. (v) In plants, the strong fibres of the cell wall are made of .....

Answer: Cellulose. The strong fibres of the plant cell wall are made of cellulose.

Question: 2. (vi) The cell membrane is made of .....

Answer: Phospholipid bilayer and proteins. The cell membrane is made of a phospholipid bilayer and proteins.

Question: 2. (vii) Diffusion is a ..... process in which no energy is required.

Answer: Passive. Diffusion is a passive process in which no energy is spent.

Question: 2. (viii) When a plant cell loses water, the cytoplasm ....., and this process is called .....

Answer: Shrinks, plasmolysis. When a plant cell loses water, the cytoplasm shrinks; this process is called plasmolysis.

Question: 2. (ix) A special type of diffusion of specific substances through carrier proteins is called .....

Answer: Facilitated diffusion. The movement of specific substances through carrier proteins is called facilitated diffusion.

Question: 2. (x) The microscopic tubules arranged in a special pattern to form centrioles are ..... in number.

Answer: 27. A centriole has 27 microscopic tubules (9 groups of three).

Question: 3. (i) What is exocytosis?

Answer: Exocytosis is the process in which things from inside the cell are pushed out of the cell by means of vesicles.

Question: 3. (ii) What are vesicles?

Answer: Vesicles are small sacs in the cell in which substances are enclosed and carried from one place to another.

Question: 3. (iv) What is nucleoplasm?

Answer: Nucleoplasm is the fluid inside the nucleus in which chromosomes

and the nucleolus are present.

Question: 3. (vi) What is plasmolysis?

Answer: Plasmolysis is the process in which a plant cell loses water and the cytoplasm shrinks.

Question: 3. (vii) What is resolution?

Answer: Resolution is the smallest distance between two close points in a microscope that can still be seen separately.

Question: 3. (viii) What is a tissue?

Answer: A tissue is a group of cells of the same kind which together do a particular job.

Question: 3. (ix) What is magnification?

Answer: Magnification is making the image of an object larger than the original in a microscope.

Question: 3. (x) What are cisternae?

Answer: Cisternae are the flattened sacs of the endoplasmic reticulum and the Golgi body.

Question: 4. (i) State the difference between prokaryotic and eukaryotic cells.

Answer: A prokaryotic cell has no true nucleus and no membrane-bound organelles. A eukaryotic cell has a true nucleus and membrane-bound organelles.

Question: 4. (ii) State the difference between mitochondria and chloroplast.

Answer: Mitochondria produce energy (respiration) and are present in all eukaryotic cells. Chloroplasts make food with light energy (photosynthesis) and are present only in plant cells.

Question: 4. (iii) State the difference between lysosome and ribosome.

Answer: Lysosomes contain digestive enzymes which break things down.

Ribosomes make proteins.

Question: 5. (i) Why is the mitochondrion called the powerhouse of the cell?

Answer: The mitochondrion is called the powerhouse of the cell because in it energy is produced in the form of ATP through respiration.

Question: 5. (ii) Why is iodine used for onion peel?

Answer: Iodine is used for onion peel because iodine stains the nucleus and other parts, so they can be seen clearly in the microscope.

Question: 5. (iii) How is the electron microscope different from the simple compound microscope?

Answer: In the electron microscope an electron beam is used and its resolution and magnification are much greater. In the simple compound microscope light is used and its magnification is about 1000 times.

Question: 5. (iv) Why is the cell membrane semi-permeable in nature?

Answer: The cell membrane is semi-permeable because some substances can pass through it and some cannot, which keeps the inside of the cell controlled.

Question: 5. (v) How is facilitated diffusion different from active transport?

Answer: In facilitated diffusion substances move from higher to lower concentration through carrier proteins and no energy is spent. In active transport substances move from lower to higher concentration and energy is spent.

Question: 5. (vi) Why is the cell of organisms called the structural and functional unit?

Answer: The cell is called the structural and functional unit because all organisms are made of cells and all the activities of life take place in the cell itself.

Question: 6. (i) Describe in detail the structure and functions of the

nucleus.

Answer: The nucleus is the control centre of the cell. Around it is a double membrane with pores. Inside are nucleoplasm, nucleolus and chromosomes. The nucleus controls all the activities of the cell and holds hereditary information.

Question: 6. (ii) What is a microscope? Describe its types.

Answer: A microscope is the instrument with which small things can be seen magnified. It has two main types: (1) Light microscope — uses light, magnification about 1000 times. (2) Electron microscope — uses an electron beam, magnification and resolution very much greater.

Question: 6. (iii) Explain the fluid mosaic model of the cell membrane.

Answer: According to the fluid mosaic model, the cell membrane is made of a phospholipid bilayer in which pieces of protein float, like things floating in water. Proteins and carbohydrates can change place in the membrane, so the membrane is flexible and moving.

جيوگهرڙي جو چڪر

Question: 1. (i) Which process is involved in mitosis?

Answer: (b) Growth, repair and asexual reproduction. Mitosis is involved in growth, repair and asexual reproduction.

Question: 1. (ii) What happens in metaphase of mitosis?

Answer: (a) The chromosomes of the cell line up at the equator. In metaphase the chromosomes come in a row at the equator.

Question: 1. (iii) Choose the odd one out.

Answer: (c) Telophase — cell membrane disappears. In telophase the nuclear membrane reappears, it does not disappear. The other three pairs are correct.

Question: 1. (iv) What happens during prophase of mitosis in animal cells?

Answer: (d) Separation of centrioles. In prophase the centrioles separate and move towards the poles.

Question: 1. (v) What is the sequence of changes by which a cell kills itself called?

Answer: (a) Apoptosis. Apoptosis is the process in which a cell ends itself in an organised way.

Question: 1. (vi) Choose the wrong statement about meiosis.

Answer: (d) New somatic cells are produced in cells. Meiosis produces gametes, not somatic cells.

Question: 1. (vii) What is the type of cell division in which a spore mother cell produces spores?

Answer: (c) Meiosis. Spores are produced by meiosis.

Question: 1. (viii) What is the stage of mitosis in which chromatids reach the poles and their movement stops?

Answer: (d) Telophase. In telophase the chromatids reach the poles and their movement stops.

Question: 1. (ix) What is the stage of meiosis in which centromeres divide and sister chromatids separate?

Answer: (a) Anaphase II. In anaphase II centromeres divide and sister chromatids separate and move towards the poles.

Question: 1. (x) What is the process in which homologous chromosomes fail to separate called?

Answer: (a) Non-disjunction. When homologous chromosomes do not separate, it is called non-disjunction.

Question: 2. (i) Chromosomes are thread-like structures that appear at the time of .....

Answer: Cell division. Chromosomes appear at the time of cell division.

Question: 2. (ii) The series of changes between one cell division and the next is called .....

Answer: Cell cycle. The series of changes from one division to the next is called the cell cycle.

Question: 2. (iii) The point where two chromatids are joined is called .....

Answer: Centromere. The point where two chromatids are joined is called the centromere.

Question: 2. (iv) Chromosomes in a cell that are similar in arrangement to each other are called .....

Answer: Homologous chromosomes. Chromosomes of similar arrangement are homologous chromosomes.

Question: 2. (v) Chromosomes in which one arm is very short and the other very long are called .....

Answer: Acrocentric. In an acrocentric chromosome one arm is very short and the other very long.

Question: 2. (vi) Two chromosomes that are genetically identical and copies of each other are called .....

Answer: Sister chromatids. Two chromatids that are genetically identical are sister chromatids.

Question: 2. (vii) The stage in which the cell grows and enzymes are made is called .....

Answer: Interphase. In interphase the cell grows and enzymes are made.

Question: 2. (viii) In organisms, meiosis produces .....

Answer: Gametes. In organisms, meiosis produces gametes.

Question: 2. (ix) During metaphase, homologous chromosomes arrange themselves in ..... order.

Answer: In pairs at the equator. In metaphase homologous chromosomes arrange themselves in pairs at the equator.

Question: 2. (x) Death of cells caused by external factors is called .....

Answer: Necrosis. Death of cells caused by external factors is called necrosis.

Question: 3. (i) What is pachytene?

Answer: Pachytene is a stage of prophase I of meiosis in which chromosomes thicken and crossing over takes place.

Question: 3. (ii) What is cytokinesis?

Answer: Cytokinesis is the division of the cytoplasm of the cell.

Question: 3. (iii) What is a bivalent?

Answer: A bivalent is a pair of homologous chromosomes formed in meiosis.

Question: 3. (iv) What are chiasmata?

Answer: Chiasmata are the points where homologous chromosomes meet each other and crossing over takes place.

Question: 3. (v) What are chromatids?

Answer: Chromatids are the two halves of a chromosome joined at the centromere.

Question: 3. (vi) What is diakinesis?

Answer: Diakinesis is the last stage of prophase I in which chromosomes thicken further and the nuclear membrane disappears.

Question: 3. (viii) What is necrosis?

Answer: Necrosis is the death of cells caused by external factors.

Question: 3. (x) What is a centromere?

Answer: A centromere is the part of the chromosome where two chromatids are joined to each other.

Question: 4. (i) State the difference between prophase I and prophase II.

Answer: In prophase I homologous chromosomes form pairs and crossing over takes place. In prophase II no crossing over takes place and chromosomes are single.

Question: 4. (ii) State the difference between apoptosis and necrosis.

Answer: Apoptosis is the organised self-destruction of a cell. Necrosis is the disorderly death of cells caused by external factors.

Question: 5. (i) Why is meiosis I called reduction division?

Answer: Meiosis I is called reduction division because in it the number of chromosomes becomes half.

Question: 5. (ii) Why is mitosis needed for growth?

Answer: Mitosis is needed for growth because new cells are formed by it and the body grows.

Question: 5. (iii) How does the chromosome number stay constant generation after generation?

Answer: The chromosome number stays constant because in meiosis the number becomes half and in fertilisation it becomes full again.

Question: 5. (iv) Why is interphase called the growth process?

Answer: Interphase is called the growth process because in it the cell grows, DNA is copied and preparation for division takes place.

Question: 5. (v) Why does interphase not occur between meiosis I and meiosis II?

Answer: Interphase does not occur between meiosis I and meiosis II because there is no need to copy DNA again; the chromosomes go directly into the second division.

Question: 6. (i) Describe the different stages of mitosis with suitable diagrams.

Answer: Stages of mitosis: prophase — chromosomes appear, nuclear membrane disappears. Metaphase — chromosomes come in a row at the equator. Anaphase — chromatids separate and move towards the poles. Telophase — chromosomes reach the poles, nuclear membrane forms again. Then by cytokinesis the cell divides into two parts.

Question: 6. (ii) Describe the stages of meiosis with suitable diagrams.

Answer: Meiosis has two divisions. Meiosis I: prophase I (leptotene, zygotene, pachytene, diplotene, diakinesis), metaphase I, anaphase I, telophase I. Meiosis II: prophase II, metaphase II, anaphase II, telophase II. In the end four gametes are formed with half the number of chromosomes.

انزائمس

Question: 1. (i) All of the following are properties of enzymes except one. Which is the one?

Answer: (c) The action of enzymes takes place with inhibitors. Enzymes speed up reactions, are sensitive to pH change and have an active site; inhibitors reduce their action, which is not a property of enzymes.

Question: 1. (ii) Enzymes are.

Answer: (b) Proteins in nature. Most enzymes are made of proteins.

Question: 1. (iii) Metabolic reactions are.

Answer: (c) I and II. Metabolism includes both constructive reactions (anabolism) and destructive reactions (catabolism).

Question: 1. (iv) What is the pH at which enzymes work best called?

Answer: (d) Optimum pH. Every enzyme has an optimum pH at which its work is best.

Question: 1. (v) The active site continuously changes its shape until the substrate binds to it. Which model does this statement describe?

Answer: (a) Induced fit model. In the induced fit model the active site changes its shape according to the substrate.

Question: 1. (vi) Choose the odd one out.

Answer: (a) Protein → carbohydrate. Protein does not act on carbohydrate; lipase acts on lipids and trypsin acts on protein.

Question: 1. (vii) Chemical reactions need specific conditions so that they can proceed correctly. What are those conditions?

Answer: (a) Temperature and pH. Chemical reactions need the correct temperature and pH.

Question: 1. (viii) All except one affect the action of enzymes. Which is the

one?

Answer: (c) Organic gradients. pH, amount of substrate and temperature affect the action of enzymes.

Question: 1. (ix) What happens to the rate of reaction when the temperature rises?

Answer: (a) It increases. When the temperature rises, the rate of reaction increases, until the enzyme is damaged.

Question: 1. (x) Choose the correct statement about the lock and key model.

Answer: (a) Enzyme and substrate have specially matching geometric shapes. In the lock and key model, enzyme and substrate fit each other like a lock and a key.

Question: 2. (i) Metabolic processes are of ..... types.

Answer: Two. Metabolism has two types: constructive (anabolism) and destructive (catabolism).

Question: 2. (ii) Enzymes ..... the activation energy of a chemical reaction.

Answer: Lower. Enzymes lower the activation energy of a reaction.

Question: 2. (iii) The presence of enzymes has no effect on the ..... of a reaction.

Answer: Equilibrium. Enzymes have no effect on the equilibrium of a reaction, they only speed up its rate.

Question: 2. (iv) In anabolic reactions, ... molecules are formed.

Answer: Large. In anabolic reactions large molecules are formed from small molecules.

Question: 2. (v) Enzyme action can be increased by .....

Answer: Activators. Activators increase the action of enzymes.

Question: 2. (vi) The small part of the enzyme where the substrate binds is

called .....

Answer: Active site. The substrate binds at the active site of the enzyme.

Question: 2. (viii) As the temperature rises, the initial reaction rate .....

Answer: Increases. As the temperature rises, the initial reaction rate increases.

Question: 2. (ix) Extreme change in pH can cause ..... of enzymes.

Answer: Denaturation. Extreme change in pH can cause denaturation of enzymes.

Question: 2. (x) In the human body there are more than ..... enzymes.

Answer: 1000. In the human body there are more than 1000 enzymes.

Question: 3. (i) What is a substrate?

Answer: A substrate is the substance on which an enzyme acts.

Question: 3. (ii) What is the active site?

Answer: The active site is the part of the enzyme where the substrate binds.

Question: 3. (iii) What are inhibitors?

Answer: Inhibitors are substances that stop or reduce the action of enzymes.

Question: 3. (iv) What are activators?

Answer: Activators are substances that increase the action of enzymes.

Question: 3. (v) What is anabolism?

Answer: Anabolism is the process in which large molecules are formed from small molecules.

Question: 3. (vi) What is catabolism?

Answer: Catabolism is the process in which large molecules break down into small molecules.

Question: 3. (vii) What is activation energy?

Answer: Activation energy is the minimum energy needed to start a

reaction.

Question: 3. (viii) What is a cofactor?

Answer: A cofactor is the non-protein part of an enzyme which is necessary for the action of the enzyme.

Question: 3. (ix) What is a prosthetic group?

Answer: A prosthetic group is a cofactor which is firmly attached to the enzyme.

Question: 3. (x) What are holoenzymes?

Answer: Holoenzymes are complete enzymes formed by the combination of apoenzyme and cofactor.

Question: 4. (i) State the difference between activator and inhibitor.

Answer: Activators increase the action of enzymes. Inhibitors stop or reduce the action of enzymes.

Question: 4. (ii) State the difference between anabolism and catabolism.

Answer: In anabolism large molecules are formed from small molecules and energy is spent. In catabolism large molecules break down into small ones and energy is released.

Question: 5. (i) Why are enzymes specific in nature?

Answer: Enzymes are specific in nature because the active site of each enzyme fits only one particular substrate.

Question: 5. (ii) How do enzymes lower the activation energy?

Answer: Enzymes bind with the substrate and lower the energy needed to start the reaction, so the reaction happens quickly.

Question: 5. (iii) Why does the presence of enzymes affect the end product?

Answer: The presence of enzymes does not affect the amount of the end product, it only speeds up the rate of the reaction; the enzyme itself is not changed at the end of the reaction.

Question: 5. (iv) How does substrate concentration affect enzyme action?

Answer: As the amount of substrate increases, the action of the enzyme speeds up, until all the enzymes are busy; after that the rate does not increase further.

Question: 5. (v) How are enzymes used in factories?

Answer: In factories enzymes are used in making paper, bread, beverages and washing products.

Question: 6. (i) What are enzymes? Describe the properties of enzymes.

Answer: Enzymes are biological catalysts that speed up chemical reactions in the cell. Properties: (1) Mostly made of proteins. (2) Act on a specific substrate. (3) Are not changed at the end of the reaction. (4) A small amount is enough. (5) Are sensitive to temperature and pH. (6) Have no effect on the equilibrium of a reaction.

Question: 6. (ii) Describe the factors affecting the action of enzymes.

Answer: Factors affecting enzyme action: (1) Temperature — action is best at suitable temperature; at very high temperature the enzyme is damaged. (2) pH — every enzyme has a suitable pH. (3) Amount of substrate — as the amount increases the rate increases until the enzymes are busy. (4) Activators and inhibitors — activators increase action, inhibitors reduce it.

حياتياتي توانائي

Question: 1. (i) In the aerobic way, 14135 kJ of energy is released. How many moles of glucose are used for this?

Answer: (c) 5. One mole of glucose gives 2827 kJ of energy; for 14135 kJ, 5 moles of glucose are used.

Question: 1. (ii) What is the stage of aerobic respiration that takes place on the cristae of mitochondria called?

Answer: (a) Electron transport chain. The electron transport chain takes place on the cristae of mitochondria.

Question: 1. (iii) In the cellular respiration process 180 ATP molecules are formed. How many moles of glucose are used during this process?

Answer: (c) 5. One mole of glucose gives 36 ATP; for 180 ATP, 5 moles of glucose are used.

Question: 1. (iv) What is the loss of electrons and protons called?

Answer: (a) Only I (oxidation reaction). The loss of electrons and protons is called oxidation.

Question: 1. (v) How much energy does each mole of ATP store?

Answer: (a) 7.3 Kcal/mole. One mole of ATP stores 7.3 Kcal of energy.

Question: 1. (vi) What is the basic molecule formed during photosynthesis called?

Answer: (a) Glucose. In photosynthesis, glucose is basically formed.

Question: 1. (vii) Where does the light-dependent reaction take place?

Answer: (b) Thylakoid. The light reaction takes place in the thylakoid of the chloroplast.

Question: 1. (viii) What is the reaction in which light energy is converted into ATP and NADPH2 called?

Answer: (a) Only I (light reaction). In the light reaction, light energy is converted into ATP and NADPH2.

Question: 1. (ix) What is the breaking of water in the presence of light called?

Answer: (c) Photolysis. The breaking of water in light is called photolysis.

Question: 1. (x) During the breaking of glucose (glycolysis), a small amount of energy is released, enough to produce.

Answer: (a) 2 ATP. In glycolysis a net of 2 ATP are formed.

Question: 2. (i) The source of energy is only .....

Answer: The sun. The original source of energy is the sun.

Question: 2. (ii) The conversion of free energy into different forms in organisms is called .....

Answer: Bioenergetics. The conversion of energy from one form to another in organisms is called bioenergetics.

Question: 2. (iii) Energy in organisms is stored in a special molecule called .....

Answer: ATP. Energy in organisms is stored in ATP.

Question: 2. (iv) Plants use ..... to make use of simple inorganic molecules, water and carbon dioxide.

Answer: Light energy. Plants use light energy to make food from water and carbon dioxide.

Question: 2. (vi) Photosynthesis is the only process that ..... free oxygen.

Answer: Produces. Only photosynthesis produces free oxygen.

Question: 2. (vii) The chloroplast is a double-membrane organelle which contains ..... semi-fluid protein.

Answer: Stroma. The semi-fluid protein liquid in the chloroplast is the stroma.

Question: 2. (viii) Different pigments in the chloroplast ..... different

light.

Answer: Absorb. Different pigments in the chloroplast absorb different light.

Question: 2. (ix) The energy obtained by breaking down a food molecule is called .....

Answer: ATP energy. The energy obtained by breaking down a food molecule is stored in the form of ATP.

Question: 2. (x) The maximum energy obtained from one mole of glucose is .....

Answer: 2827 kJ/mole. One mole of glucose gives 2827 kJ of energy.

Question: 3. (i) What is bioenergetics?

Answer: Bioenergetics is the science in which the conversion and use of energy in organisms is studied.

Question: 3. (ii) What is energy?

Answer: Energy is the ability to do work.

Question: 3. (iii) What is oxidation?

Answer: Oxidation is the reaction in which electrons and protons are lost.

Question: 3. (iv) What is a food chain?

Answer: A food chain is the line of organisms in which one organism eats another and energy moves forward.

Question: 3. (v) What are grana?

Answer: Grana are stacks of thylakoids in the chloroplast.

Question: 3. (vi) What is photolysis?

Answer: Photolysis is the breaking of water in light, from which oxygen, protons and electrons come out.

Question: 3. (vii) What is fermentation?

Answer: Fermentation is the process in which glucose breaks down without oxygen and a little energy is obtained.

Question: 3. (viii) What is stroma?

Answer: Stroma is the semi-fluid protein liquid of the chloroplast in which grana float.

Question: 3. (ix) What is pyruvic acid?

Answer: Pyruvic acid is the substance formed at the end of glycolysis which goes into the Krebs cycle.

Question: 3. (x) What is aerobic respiration?

Answer: Aerobic respiration is the process in which glucose breaks down in the presence of oxygen and a lot of energy (36 ATP) is obtained.

Question: 4. (i) State the difference between respiration and photosynthesis.

Answer: In respiration glucose breaks down and energy is released; in photosynthesis glucose is made with light energy. In respiration oxygen is used and carbon dioxide comes out; in photosynthesis the reverse happens.

Question: 4. (ii) State the difference between light reaction and dark reaction.

Answer: The light reaction takes place in the thylakoid and needs light; in it ATP and NADPH2 are formed. The dark reaction takes place in the stroma and does not need direct light; in it glucose is formed.

Question: 4. (iii) State the difference between aerobic and anaerobic respiration.

Answer: In aerobic respiration oxygen is used and 36 ATP are obtained. In anaerobic respiration oxygen is not used and only 2 ATP are obtained.

Question: 5. (i) How do carbon dioxide and temperature affect the rate of photosynthesis?

Answer: As carbon dioxide increases, the rate of photosynthesis increases, until other factors run short. As temperature increases the rate increases,

but at very high temperature enzymes are damaged and the rate decreases.

Question: 5. (ii) Why is the second stage of photosynthesis called the dark reaction?

Answer: The second stage of photosynthesis is called the dark reaction because it does not need direct light; it uses the ATP and NADPH2 obtained from the light reaction.

Question: 5. (iii) How are breathing and cellular respiration different from each other?

Answer: Breathing is the physical process of air coming in and going out. Cellular respiration is the chemical process of the breaking down of glucose in the cell, by which energy is obtained.

Question: 5. (iv) What is the harm of lactic acid fermentation?

Answer: In lactic acid fermentation, lactic acid collects, which causes tiredness and pain in the muscles.

Question: 5. (v) How is glucose converted into secondary products in plants?

Answer: In plants glucose is converted into starch, cellulose, sucrose and other substances.

Question: 6. (i) What is the treasure of cellular energy? Describe the method of transfer of chemical energy.

Answer: The treasure of cellular energy is ATP. Chemical energy is stored in food molecules; in respiration that energy is transferred into ATP and then used in the activities of the cell.

Question: 6. (ii) Describe the stages of photosynthesis with a suitable diagram.

Answer: Photosynthesis has two stages: (1) Light reaction — takes place in the thylakoid; water breaks (photolysis) and ATP and NADPH2 are formed.

(2) Dark reaction — takes place in the stroma; carbon dioxide is converted into glucose with the help of ATP and NADPH2 (Calvin cycle).

Question: 6. (iii) Describe the process of aerobic respiration in living organisms.

Answer: Aerobic respiration has three stages: (1) Glycolysis — in the cytosol glucose breaks into pyruvic acid, 2 ATP are obtained. (2) Krebs cycle — in the mitochondria pyruvic acid breaks down and carbon dioxide comes out. (3) Electron transport chain — ATP are formed on the cristae. A total of 36 ATP are obtained.

غذائیت

Question: 1. (i) Choose the mismatched pair.

Answer: (c) Fat → starch. Fat breaks into fatty acids and glycerol, not starch; the other three pairs are correct.

Question: 1. (ii) Which disease is caused by deficiency of vitamin C?

Answer: (c) Scurvy. Deficiency of vitamin C causes scurvy.

Question: 1. (iii) What is the period of rapid growth and maturity called?

Answer: (c) Puberty. The period of rapid growth and maturity is puberty.

Question: 1. (iv) Protein deficiency in nutrition creates imbalance in the body's system. Which disease is this?

Answer: (d) Kwashiorkor. Protein deficiency causes kwashiorkor.

Question: 1. (v) An animal that covers a long distance in migration benefits most from storing energy in which form?

Answer: (a) Fat. Fat stores the most energy, so migrating animals store energy in the form of fat.

Question: 1. (vi) Which of the following vitamins are correctly matched with their use?

Answer: (d) Vitamin A → is involved in the eye's vision pigment. Vitamin A is necessary for the vision pigment (rhodopsin).

Question: 1. (vii) Which of the following statements explains pepsin?

Answer: (d) It starts the digestion of protein in the stomach. Pepsin starts the digestion of protein in the stomach.

Question: 1. (viii) Which statement is correct about bile (the fat-dissolving secretion)?

Answer: (c) It emulsifies fat in the intestine. Bile divides fat into small drops so that enzymes can act on it.

Question: 1. (ix) In the human digestive system, where do the trachea and oesophagus both meet?

Answer: (c) In the pharynx. The trachea and oesophagus both meet in the pharynx.

Question: 1. (x) All are sources of calcium except one. Which is the one?

Answer: (a) Red meat. Green leafy vegetables, broccoli and walnuts are sources of calcium; meat is not a good source of calcium.

Question: 2. (i) Seafood is a very good source of protein because it is often low in .....

Answer: Fat. Seafood is high in protein and low in fat.

Question: 2. (ii) Iron deficiency is related to low immunity and .....

Answer: Anaemia. Iron deficiency lowers immunity and causes anaemia.

Question: 2. (iii) The rhythmic and sequential movement of muscles is called .....

Answer: Peristalsis. The sequential movement of muscles is called peristalsis.

Question: 2. (iv) The juices collected from the pancreas contain different .....

Answer: Enzymes. Pancreatic juice contains different enzymes, such as lipase, amylase and trypsin.

Question: 2. (v) Organisms that obtain their food from decayed matter are called .....

Answer: Saprophytic. Organisms that obtain food from decayed matter are called saprophytic.

Question: 2. (vi) Food that fulfils all the needs of the body is called .....

Answer: Balanced diet. Food that fulfils all needs is called a balanced diet.

Question: 2. (vii) Fertiliser is a substance in which the ..... necessary for the food of plants are present.

Answer: Nutrients. Fertiliser contains the nutrients necessary for plants.

Question: 2. (viii) Excessive use of fertilisers ..... the ecosystem.

Answer: Damages. Excessive use of fertilisers damages the environment.

Question: 2. (ix) Carbohydrates that are digested quickly are called .....

Answer: Simple carbohydrates. Carbohydrates that are digested quickly are called simple carbohydrates.

Question: 2. (x) Each gram of fat gives the body about .....

Answer: 9 calories. Each gram of fat gives about 9 calories.

Question: 3. (i) What are vitamins?

Answer: Vitamins are organic substances needed by the body in small amounts and necessary for health.

Question: 3. (ii) What is malnutrition?

Answer: Malnutrition is food in which the nutrients needed by the body are not obtained in full amount.

Question: 3. (iii) What is goitre?

Answer: Goitre is the swelling of the thyroid gland in the neck, caused by deficiency of iodine.

Question: 3. (iv) What is anaemia?

Answer: Anaemia is the condition in which the amount of red blood cells or haemoglobin in the blood decreases.

Question: 3. (v) What is constipation?

Answer: Constipation is the condition in which the belly is not cleared and waste comes out with difficulty.

Question: 3. (vi) What is obesity?

Answer: Obesity is the condition in which too much fat collects in the body.

Question: 3. (vii) What is starvation?

Answer: Starvation is the condition in which the body does not get the

food it needs and the body becomes weak.

Question: 3. (viii) What is digestion?

Answer: Digestion is the process in which food breaks into small molecules so that the body can absorb it.

Question: 3. (ix) What is chyme?

Answer: Chyme is the semi-liquid substance of half-digested food in the stomach.

Question: 3. (x) What is an ulcer?

Answer: An ulcer is a wound in the inner wall of the stomach or intestines.

Question: 4. (i) State the difference between fat-soluble and water-soluble vitamins.

Answer: Fat-soluble vitamins (A, D, E, K) dissolve in fat and are stored in the body. Water-soluble vitamins (B and C) dissolve in water and are not stored in the body.

Question: 4. (ii) State the difference between marasmus and kwashiorkor.

Answer: Marasmus is caused by deficiency of energy (calories) and the child becomes very weak. Kwashiorkor is caused by deficiency of protein and the body swells.

Question: 4. (iii) State the difference between chemical and mechanical digestion.

Answer: In mechanical digestion food is broken into small pieces by chewing and peristalsis. In chemical digestion enzymes break food down chemically.

Question: 4. (iv) State the difference between micronutrients and macronutrients.

Answer: Micronutrients are nutrients needed by the body in small amounts, such as vitamins and minerals. Macronutrients are those needed in large amounts, such as carbohydrates, proteins and fats.

Question: 4. (v) State the difference between autotrophic and heterotrophic nutrition.

Answer: In autotrophic nutrition organisms make their own food, such as plants. In heterotrophic nutrition organisms depend on others for food, such as animals.

Question: 4. (vi) State the difference between inorganic and organic fertilisers.

Answer: Inorganic fertilisers are made in chemical factories and act quickly, but harm the environment. Organic fertilisers are made from natural materials, such as dung, and improve the soil.

Question: 5. (i) Why are fertilisers necessary for plants?

Answer: Fertilisers are necessary for plants because from them plants get nitrogen, phosphorus and other nutrients with which plants grow.

Question: 5. (ii) How does the stomach protect itself from its acidic environment?

Answer: The stomach protects itself from its acidic environment with a thick layer of mucus which covers the wall of the stomach.

Question: 5. (iii) Why is nitrogen needed for plants?

Answer: Nitrogen is needed for plants because it is required for making proteins and chlorophyll.

Question: 5. (iv) Why is fat called a more powerful food?

Answer: Fat is called a more powerful food because each gram of it gives about 9 calories, which is double that of carbohydrates and proteins.

Question: 5. (v) Why is chewing necessary for swallowing?

Answer: Chewing is necessary because food becomes small pieces, mixes with saliva, and swallowing and digestion become easy.

Question: 6. (i) Describe the environmental hazards related to chemical fertilisers.

Answer: Excessive use of chemical fertilisers damages the soil, pollutes water, increases algae in lakes (eutrophication) and increases greenhouse gas in the air. That is why chemical fertilisers are dangerous for the environment.

Question: 6. (ii) Describe the action of the stomach and the intestine in the human digestive system, with a suitable diagram.

Answer: In the stomach food turns into chyme; pepsin breaks protein and acid kills germs. In the small intestine pancreatic juice, bile and intestinal juice break food down completely, and it is absorbed through the villi.

Question: 6. (iii) What are vitamins? Describe the types of vitamins.

Answer: Vitamins are organic substances needed by the body in small amounts. There are two types: (1) Fat-soluble — A, D, E, K. (2) Water-soluble — B and C.

Question: 6. (iv) Describe mineral-deficiency diseases in humans.

Answer: Deficiency of minerals causes diseases: deficiency of iron causes anaemia, deficiency of iodine causes goitre, deficiency of calcium makes bones weak and teeth bad.

Question: 6. (v) Describe the effects of malnutrition on humans.

Answer: Effects of malnutrition: growth stops in children, kwashiorkor and marasmus occur, immunity decreases, anaemia occurs, heart diseases, constipation and obesity occur.

منتقلي

Question: 1. (i) Through what are organic substances (food) transported?

Answer: (d) Phloem. Organic substances (food) are transported through phloem.

Question: 1. (ii) By what arrangement does water move up through the xylem?

Answer: (c) Transpiration of water from the leaf. Water rises in the xylem by the pull of transpiration.

Question: 1. (iii) Phloem transport of sucrose can be described as the moving mechanism from "source" to "sink". What is the "sink"?

Answer: (d) Both (b) and (c). The sink can be both the storage organ and the growing part.

Question: 1. (iv) Human plasma proteins include which of the following?

Answer: (d) I and III. Plasma proteins include fibrinogen and albumin; haemoglobin is not in plasma but in red blood cells.

Question: 1. (vi) Where does deoxygenated blood returning to the human heart first flow?

Answer: (a) Into the right atrium. Deoxygenated blood first comes into the right atrium.

Question: 1. (vii) Root hairs are very important because they .....

Answer: (d) Increase the absorbing surface. Root hairs increase the surface for absorbing water and minerals.

Question: 1. (viii) What is the type of circulatory system in which blood flows through vessels called?

Answer: (b) Only II (closed circulatory system). When blood flows in vessels it is called a closed circulatory system.

Question: 1. (ix) What is the part between endodermis and epidermis called?

Answer: (c) Cortex. Between the endodermis and epidermis is the cortex.

Question: 1. (x) What is the movement of water molecules from higher to lower concentration called?

Answer: (b) Osmosis. The movement of water molecules from higher to lower concentration is called osmosis.

Question: 2. (i) Loss of water from the inner parts of plants in the form of vapour is called .....

Answer: Transpiration. Loss of water from plants in the form of vapour is called transpiration.

Question: 2. (ii) In flowering plants, xylem is made of ..... types of tissues.

Answer: Four. In flowering plants xylem is made of four types of tissues: tracheids, vessels, xylem parenchyma and xylem fibres.

Question: 2. (iii) Root hairs increase the ..... for absorption.

Answer: Surface area. Root hairs increase the surface area for absorption.

Question: 2. (iv) Opening and closing of stomata occurs with .....

Answer: Guard cells. Opening and closing of stomata occurs with guard cells.

Question: 2. (v) The type of cancer affecting the bone marrow is called .....

Answer: Leukaemia. Cancer of the bone marrow is called leukaemia.

Question: 2. (vi) The rhythmic pumping of blood by the heart throughout the body is called .....

Answer: Heartbeat (pulse). The rhythmic pumping of the heart is called heartbeat.

Question: 3. (i) What is blood?

Answer: Blood is a red fluid which transports food, oxygen and waste in the body.

Question: 3. (ii) What is water loss?

Answer: Water loss is the loss of water from plants in the form of vapour (transpiration).

Question: 3. (iii) What is osmosis?

Answer: Osmosis is the movement of water molecules through a semi-permeable membrane from higher to lower concentration.

Question: 3. (iv) What are stomata?

Answer: Stomata are small pores on the surface of leaves through which gases and water vapour pass.

Question: 3. (v) What is a dorsiventral leaf?

Answer: A dorsiventral leaf is a leaf in which there is a difference between the upper and lower surfaces.

Question: 3. (vii) What is a sieve plate?

Answer: A sieve plate is the wall with pores between the tubes of phloem through which food passes.

Question: 3. (viii) What is a sink?

Answer: A sink is the part of the plant where food is stored or used, such as a storage organ or a growing part.

Question: 3. (ix) What are granulocytes?

Answer: Granulocytes are a type of white blood cells which contain granules.

Question: 3. (x) What is blood pressure?

Answer: Blood pressure is the pressure which blood exerts on the walls of the vessels.

Question: 4. (i) State the difference between pulmonary and systemic circuits.

Answer: In the pulmonary circuit blood goes from the heart to the lungs and from the lungs to the heart. In the systemic circuit blood goes from the heart to all the organs of the body and back to the heart.

Question: 4. (ii) State the difference between open and closed circulatory systems.

Answer: In an open circulatory system blood flows in open spaces outside vessels. In a closed circulatory system blood always stays in vessels.

Question: 4. (iii) State the difference between xylem and phloem.

Answer: Xylem carries water and minerals from the roots to the leaves. Phloem carries food (sucrose) from the leaves to other parts of the plant.

Question: 4. (iv) State the difference between arteries and veins.

Answer: Arteries carry blood from the heart to the organs and their walls are thick. Veins carry blood from the organs to the heart and their walls are thin.

Question: 4. (v) State the difference between RBC and WBC.

Answer: Red blood cells (RBC) carry oxygen and have no nucleus. White blood cells (WBC) protect against diseases and have a nucleus.

Question: 5. (i) Why are capillaries made of a single layer of endothelium?

Answer: Capillaries are made of a single layer of endothelium so that substances can easily pass through them.

Question: 5. (ii) Why do plants need transpiration?

Answer: Plants need transpiration because by it water and minerals rise up, the plant stays cool, and food is transported.

Question: 5. (iii) How does water move in the xylem?

Answer: Water rises in the xylem by the pull of transpiration, the sticking of water molecules, and root pressure.

Question: 5. (iv) What are semilunar valves?

Answer: Semilunar valves are valves that let blood flow in one direction

only and stop it from flowing back.

Question: 5. (v) How does atherosclerosis cause stroke and heart attack?

Answer: In atherosclerosis fat collects in the vessels and the flow of blood stops. A blockage in the brain causes a stroke, and a blockage in the heart causes a heart attack.

Question: 6. (i) Describe the structure of the heart with a suitable diagram.

Answer: The heart is divided into four chambers: right atrium, right ventricle, left atrium, left ventricle. It has tricuspid, bicuspid and semilunar valves. The heart pumps blood and is protected in the pericardium.

Question: 6. (ii) What is blood? Describe the function of the composition of blood.

Answer: Blood is a red fluid. It has two main parts: plasma (55%) — which contains water, proteins and nutrients; and corpuscles — red blood cells (carry oxygen), white blood cells (protect), and platelets (clot the blood).

Question: 6. (iii) What is transpiration? Describe its mechanism and the factors affecting it.

Answer: Transpiration is the loss of water from plants in the form of vapour. Mechanism: stomata open, water evaporates, and a pull is created in the xylem. Factors affecting it: temperature, humidity, wind and atmospheric pressure.